

Pentagonal Number Theorem

JC PM MV

April 28, 2026

1 Partitions

Definition 1. A *partition* of a positive integer n , also called an integer partition, is a way of writing n as a sum of positive integers. Two sums that differ only in the order of their summands are considered the same partition.

The *partition function* $p(n)$ represents the number of possible partitions of a natural number n .

Example 2. $5 = 4 + 1 = 3 + 2 = 3 + 1 + 1 = 2 + 2 + 1 = 2 + 1 + 1 + 1 = 1 + 1 + 1 + 1 + 1$
 $p(5) = 7$

Lemma 3. The generating function for $p(n)$ is given by

$$\sum_{n=0}^{\infty} p(n) x^n = \prod_{k=1}^{\infty} (1 + x^k + x^{2k} + \dots) = \prod_{k=1}^{\infty} (1 - x^k)^{-1}.$$

Definition 4 (Partitions of n into distinct parts). Let $n \in \mathbb{N}$. We define the following sets of partitions of n into distinct positive parts:

$$\begin{aligned} \mathcal{P}(n) &= \left\{ S \subseteq \{1, \dots, n\} \mid \sum_{s \in S} s = n \right\}, \\ \mathcal{P}_{\text{even}}(n) &= \{S \in \mathcal{P}(n) \mid |S| \equiv 0 \pmod{2}\}, \\ \mathcal{P}_{\text{odd}}(n) &= \{S \in \mathcal{P}(n) \mid |S| \equiv 1 \pmod{2}\}. \end{aligned}$$

We further define the number of even and odd partitions of n as $p_e(n) = |\mathcal{P}_{\text{even}}(n)|$ and $p_o(n) = |\mathcal{P}_{\text{odd}}(n)|$.

Lemma 5. We have

$$\prod_{k=1}^{\infty} (1 - x^k) = \sum_{s=0}^{\infty} \sum_{k_1 < k_2 < \dots < k_s} (-1)^s x^{k_1 + k_2 + \dots + k_s} = \sum_{n=0}^{\infty} c_n x^n$$

where $c_0 = 1$ and for $n \geq 1$ and $c_n = p_e(n) - p_o(n)$.

Example 6. We compute $\prod_{i=1}^{20} (1 - x^i) + o(x^{20})$.

The answer is $1 - x - x^2 + x^5 + x^7 - x^{12} - x^{15} + O(x^{20})$.

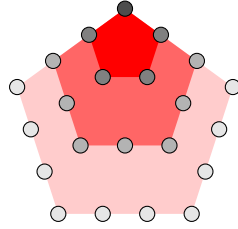
n		$p_o(n)$	$p_e(n)$
1	1	1	0
2	2	1	0
3	$3 = 2 + 1$	1	1
4	$4 = 3 + 1$	1	1
5	$5 = 4 + 1 = 3 + 2$	1	2
6	$6 = 5 + 1 = 4 + 2 = 3 + 2 + 1$	2	2
7	$7 = 6 + 1 = 5 + 2 = 4 + 3 = 4 + 2 + 1$	2	3

2 Pentagonal number theorem

Theorem 7 (Euler).

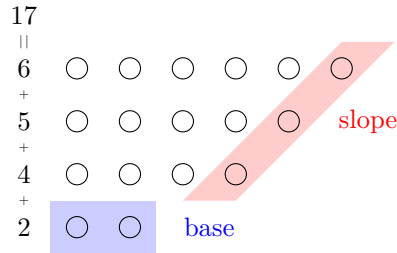
$$\prod_{i=1}^{\infty} (1 - x^i) = 1 + \sum_{k=1}^{\infty} (-1)^k (x^{(3k^2-k)/2} + x^{(3k^2+k)/2}) = \sum_{k \in \mathbb{Z}} (-1)^k x^{(3k^2-k)/2}.$$

Remark 8 (Pentagonal numbers). $\frac{k}{\frac{3k^2-k}{2}} \left| \begin{array}{c} -3 \\ 15 \end{array} \right| \begin{array}{c} -2 \\ 7 \end{array} \left| \begin{array}{c} -1 \\ 2 \end{array} \right| \begin{array}{c} 0 \\ 0 \end{array} \left| \begin{array}{c} 1 \\ 1 \end{array} \right| \begin{array}{c} 2 \\ 5 \end{array} \left| \begin{array}{c} 3 \\ 12 \end{array} \right|$



2.1 Proof

Consider a diagram of a partition of n into different parts.



Definition 9 (Base and slope). Consider a partition $S \subseteq \{1, \dots, n\}$ of n into different parts. We define the *base* $b \in S$ as the smallest element of the partition. Let $\max(S)$ denote the largest element of S . We define the *slope* s as:

$$s = \max\{k \geq 1 : \max(S), \max(S) - 1, \max(S) - 2, \dots, \max(S) - k + 1 \in S\}.$$

and the *slope set* D as $\{\max(S), \max(S) - 1, \dots, \max(S) - s + 1\}$.

Definition 10. Let $n \in \mathbb{N}$. We define the following sets of partitions of n into distinct positive parts:

$$\begin{aligned}\mathcal{P}_\alpha(n) &= \{S \in \mathcal{P}(n) \mid b \leq s \text{ and } b \notin D, \text{ or if } b \leq s - 1\}, \\ \mathcal{P}_\beta(n) &= \{S \in \mathcal{P}(n) \mid b > s \text{ and } b \notin D, \text{ or if } b \geq s + 2\}, \\ \mathcal{P}_{\text{special}}(n) &= \{S \in \mathcal{P}(n) \mid b \in D \text{ and } |b| = |s| \text{ or } |b| = |s| + 1\}.\end{aligned}$$

Lemma 11. For all natural numbers n the following holds: The sets $\mathcal{P}_\alpha(n)$, $\mathcal{P}_\beta(n)$, $\mathcal{P}_{\text{special}}(n)$ are pairwise disjoint.

Lemma 12. For all natural numbers n the following holds

$$\mathcal{P}(n) = \mathcal{P}_\alpha(n) \sqcup \mathcal{P}_\beta(n) \sqcup \mathcal{P}_{\text{special}}(n).$$

Definition 13. For $k \in \mathbb{Z}_{\geq 1}$ define $S_{-k} = \{2k - 1, 2k - 2, \dots, k\}$. For $k \in \mathbb{Z}_{\geq 0}$ define $S_k = \{2k, 2k - 1, \dots, k\}$.

Lemma 14. If n is not a pentagonal number, then $\mathcal{P}_{\text{special}}(n) = \emptyset$.

If $n = \frac{3k^2 - k}{2}$ for some $k \in \mathbb{Z}_{\geq 1}$, then $\mathcal{P}_{\text{special}}(n) = \{S_{-k}\}$.

If $n = \frac{3k^2 + k}{2}$ for some $k \in \mathbb{Z}_{\geq 0}$, then $\mathcal{P}_{\text{special}}(n) = \{S_k\}$.

Definition 15 (Operations α and β). Let $S \subseteq \{1, \dots, n\}$ be a partition of n into different parts with base b and slope s and slope set D . We define the following operations on S :

α : If $S \in \mathcal{P}_\alpha$ (equivalently, if $b \leq s$ and $b \notin D$, or if $b \leq s - 1$), then we move b up and turn its points into new slope.

$$\alpha(S) = (S \setminus \{b, \max(S) - b + 1\}) \cup \{\max(S) + 1\}.$$

β : If $S \in \mathcal{P}_\beta$ (equivalently, if $b > s$ and $b \notin D$, or if $b \geq s + 2$), then we move the slope down and turn its points into the new base.

$$\beta(S) = S \cup \{s, \max(S) - s\} \setminus \{\max(S)\}.$$

α



β



Lemma 16. If $S \in \mathcal{P}_\alpha(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n)$.

Lemma 17. If $S \in \mathcal{P}_\beta(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n)$.

Lemma 18. *The composition of maps $\beta \circ \alpha$ is equal to the identity map on $\mathcal{P}_\alpha(n)$. Equivalently, if $S \in \mathcal{P}_\alpha(n)$ then $\beta(\alpha(S)) = S$.*

Lemma 19. *The composition of maps $\alpha \circ \beta$ is equal to the identity map on $\mathcal{P}_\beta(n)$. Equivalently, if $S \in \mathcal{P}_\beta(n)$ then $\alpha(\beta(S)) = S$.*

Lemma 20. *The map $\alpha : \mathcal{P}_\alpha(n) \rightarrow \mathcal{P}_\beta(n)$ is a bijection. The map $\beta : \mathcal{P}_\beta(n) \rightarrow \mathcal{P}_\alpha(n)$ is a bijection.*

Lemma 21.

$$|\mathcal{P}_\alpha(n)| = |\mathcal{P}_\beta(n)|.$$

Lemma 22. *The operations α and β change the parity of the number of parts of a partition. More precisely,*

- *If $S \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)$;*
- *If $S \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)$;*
- *If $S \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)$;*
- *If $S \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)$.*

Lemma 23.

$$\begin{aligned} |\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)| &= |\mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)|, \\ |\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)| &= |\mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)|. \end{aligned}$$

Lemma 24.

$$p_e(n) - p_o(n) = \begin{cases} (-1)^k, & \text{if } n = \frac{3k^2 \pm k}{2} \\ 0, & \text{otherwise.} \end{cases}$$

Proof. We have

$$p_e(n) - p_o(n) = |\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)|. \quad (1)$$

By Lemma 12 we have

$$|\mathcal{P}_{\text{even}}(n)| = |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_\alpha(n)| + |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_\beta(n)| + |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)|$$

and

$$|\mathcal{P}_{\text{odd}}(n)| = |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_\alpha(n)| + |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_\beta(n)| + |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)|.$$

It follows from two previous equations and Lemma 23 that

$$|\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)| = |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)| - |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)|. \quad (2)$$

Finally, the statement of the lemma follows from the equations (1), (2) and Lemma 14. $\square$

3 Jacobi triple product

Theorem 25. *For all $w \neq 0$ and q with $|q| < 1$:*

$$\prod_{k=1}^{\infty} (1 - q^{2k}) (1 - q^{2k-1} w^2) (1 - q^{2k-1} w^{-2}) = \sum_{k=-\infty}^{\infty} (-1)^k w^{2k} q^{k^2}.$$

3.1 Proof

Set $\phi_n(w, q) := \prod_{k=1}^n (1 - q^{2k-1} w^2) (1 - q^{2k-1} w^{-2})$. We have

$$\phi_n(qw, q) = \phi_n(w, q) \frac{(1 - q^{2n+1} w^2) (1 - q^{-1} w^{-2})}{(1 - qw^2) (1 - q^{2n-1} w^{-2})} = \phi_n(w, q) \frac{1 - q^{2n+1} w^2}{-qw^2 + q^{2n}}. \quad (3)$$

We consider the following expansion

$$\phi_n(w, q) = \sum_{k=-n}^n A_{n,k}(q) w^{2k}, \quad (4)$$

where $A_{n,k}(q)$ is a polynomial in q . We have a symmetry $A_{n,k}(q) = A_{n,-k}(q)$. We find

$$A_{n,n}(q) = q^{1+3+5+\dots+(2n-1)} = q^{n^2}. \quad (5)$$

We substitute (4) into (3) and obtain

$$A_{n,k}(q) = A_{n,k-1}(q) q^{2k-1} \frac{1 - q^{2n-2k+2}}{1 - q^{2n+2k}}. \quad (6)$$

From (5) and (6) we find

$$A_{n,k} = \frac{q^{k^2}}{(1 - q^2) \dots (1 - q^{2n})} \prod_{s=n-k+1}^n (1 - q^{2s}) \prod_{s=n+k+1}^{2n} (1 - q^{2s}). \quad (7)$$

From (7) we obtain

$$\prod_{k=1}^n (1 - q^{2k}) (1 - q^{2k-1} w^2) (1 - q^{2k-1} w^{-2}) = \sum_{k=-n}^n (-1)^k w^{2k} q^{k^2} B_{n,k}(q).$$

where

$$B_{n,k}(q) = \prod_{s=n-k+1}^n (1 - q^{2s}) \prod_{s=n+k+1}^{2n} (1 - q^{2s}).$$

We have $B_{n,k} = 1 + O(q^{n-k+1})$. We take the limit $n \rightarrow \infty$ and finish the proof.